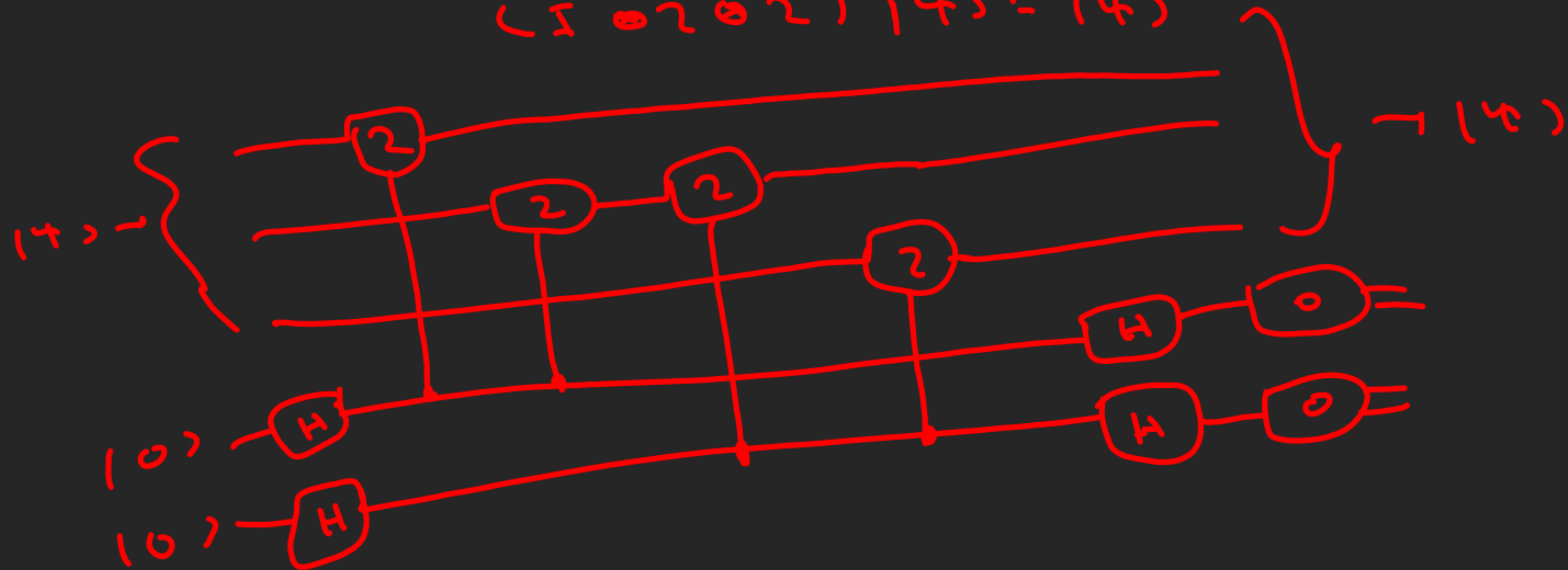


# Pauli Observables for Repetition Code

$$|4\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$(Z \otimes Z \otimes I) |4\rangle = |4\rangle$$

$$(I \otimes Z \otimes Z) |4\rangle = |4\rangle$$





# Error Detection

$$(4) \rightarrow (x \oplus 1 \oplus 1) (4)$$

↓  
Error

$$(2 \oplus 2 \oplus 1) ((x \oplus 1 \oplus 1) (4)) = - (x \oplus 1 \oplus 1) (4)$$

$$(1 \oplus 2 \oplus 2) ((x \oplus 1 \oplus 1) (4)) = (x \oplus 1 \oplus 1) (4)$$

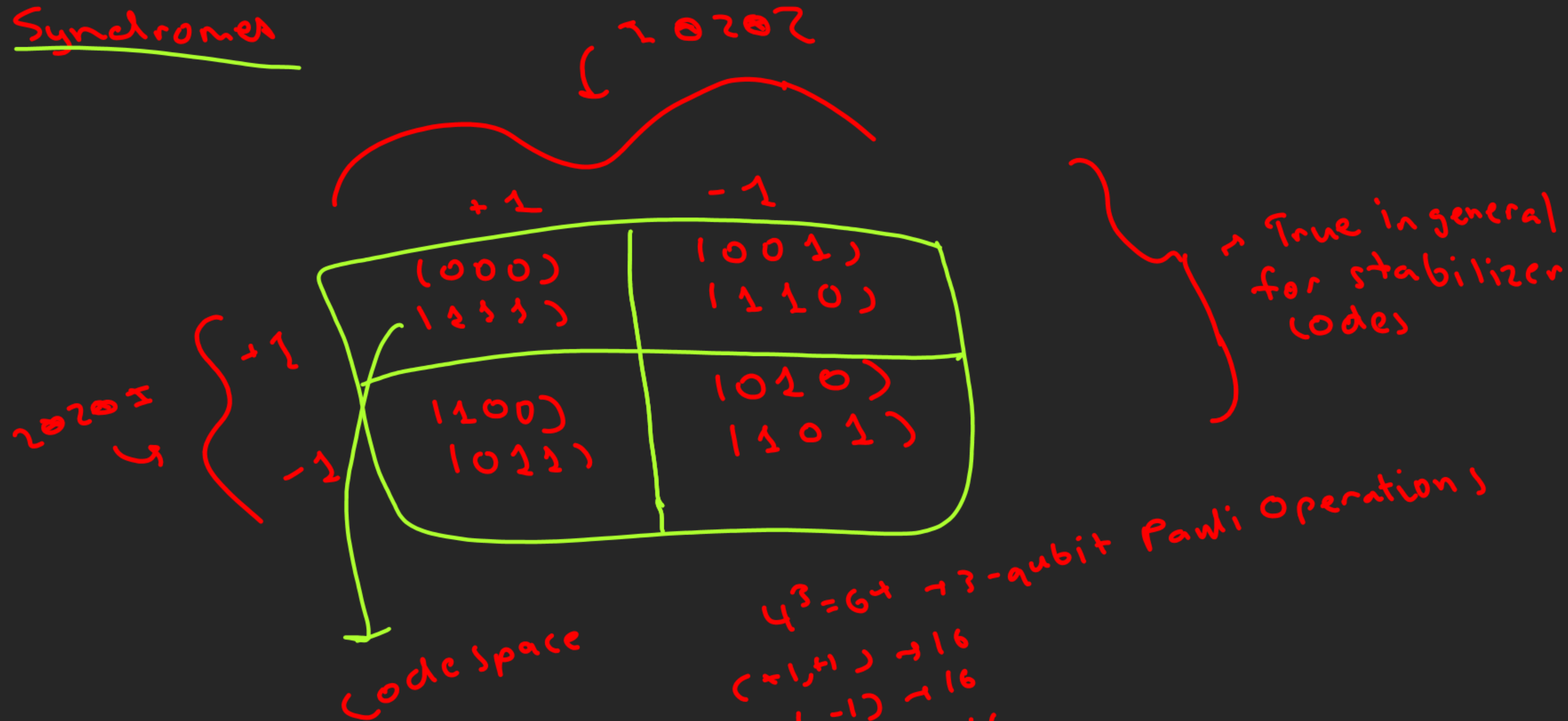
|                       | $1 \oplus 1 \oplus 1$ | $x \oplus 1 \oplus 1$ | $1 \oplus x \oplus 1$ | $1 \oplus 1 \oplus x$ |
|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| $2 \oplus 2 \oplus 1$ | +1                    | -1                    | -2                    | +1                    |
| $1 \oplus 2 \oplus 2$ | +1                    | +2                    | -1                    | -1                    |

$$\begin{aligned} +1 &\rightarrow 0 \\ -1 &\rightarrow 1 \end{aligned}$$

(columns) → Syndromes



# Syndromes



$4^3 = 64 \rightarrow 3\text{-qubit Pauli operations}$   
 $(+1, +1) \rightarrow 16$   
 $(+1, -1) \rightarrow 16$   
 $(-1, +1) \rightarrow 16$   
 $(-1, -1) \rightarrow 16$